

EGFR fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes EGFR gene fusions from the msk_impact_50k_2026 study, covering 55 fusion events. 28/40 fusions with known frame status (70.0%) were in-frame. 41/55 fusions (74.5%) retained the Protein tyrosine and serine/threonine kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0114539$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for EGFR.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 55 analyzed fusion events, identifying 31 distinct fusion partner genes. The most frequent partner was SEPTIN14, observed in 18 events.

Domain retention

Protein tyrosine and serine/threonine kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 25/28 (89.3%) of in-frame fusions retained the domain, $p=0.0114539$ (statistically significant at $\alpha=0.05$). The Protein tyrosine and serine/threonine kinase appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.227723.

Domain disruption

Disruption of the Receptor L domain, Furin-like cysteine rich region, and Growth factor receptor domain IV was tested with Fisher's exact test comparing in-frame fusions against all others; 6/28 (21.4%) of in-frame fusions disrupted the domain, $p=0.991541$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.336634.

Cutpoint detection

Cutpoint detection scanned 55 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 981 aa (exon 24; permutation-corrected $p=0.128713$, not statistically significant at $\alpha=0.05$). This is 16 aa from the nearest configured domain boundary at 965 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=41$) and "not_retained" ($n=14$). For Frame_status, retained: 25/31 (80.6%, 95% CI 63.7%-90.8%); not_retained: 3/9 (33.3%, 95% CI 12.1%-64.6%). Welch's t-test comparing Tumor_variant_count between groups gave means of 358.805 vs 214.071, $p=0.389355$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (31 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for EGFR in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (16 row(s) in genomic_bin_recurrence, 3 row(s) in exact_position_recurrence, 6 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: EGFR has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run (3 row(s) in retention_counter_intuitive_events); see the accompanying table.

Mutation cooccurrence

Mutation cooccurrence reported: EGFR has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (122 row(s) in window_scan, 64 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SEPTIN14	18	0.3272727272727272	0.06425935377650496	0.047284245674038744	0.9779763991120458	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3305926386008197	1
RAD51	2	0.03636363636363636	0.06425935377650496	0.047284245674038744	0.9989484752891693	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.24646572275466094	2
LANCL2	3	0.05454545454545454	0.06425935377650496	0.047284245674038744	0.961794602173151	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.24634718724180368	3
ELAPOR2	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	1.0	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2411689060067401	4
PKD1L1	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.9989484752891693	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.24101117730011548	5
VPS41	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.9947423764458465	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.24038026247361707	6
ZNF713	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.9400630914826499	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.23217836972913758	7
VWC2	2	0.03636363636363636	0.06425935377650496	0.047284245674038744	0.9032597266035752	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.23211241045182185	8
NIPSNAP2	3	0.05454545454545454	0.06425935377650496	0.047284245674038744	0.8647038205397827	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.23178356999679842	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TNRC18	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.9232386961093586	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2296547104231439	10
BLTP3B	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.9011566771819137	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.22634240758402716	11
CDH7	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.8338590956887487	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2162477703600524	12
FADD	2	0.03636363636363636	0.06425935377650496	0.047284245674038744	0.6908517350157728	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.20025121171365148	13
CADPS	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.6919032597266035	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19495439496573064	14
EEA1	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.6919032597266035	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19495439496573064	15
KIF5B	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.6919032597266035	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19495439496573064	16
NUMA1	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.6919032597266035	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19495439496573064	17
VSTM2A	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.6908517350157728	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19479666625910602	18
PDE1C	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.6529968454258674	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.1891184328206202	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
YIF1B	1	0.018181818181818	0.06425935377650496	0.047284245674038744	0.6414300736067298	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18738341704774958	20
TUT7	1	0.018181818181818	0.06425935377650496	0.047284245674038744	0.6277602523659306	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.18533294386162968	21
CSF2RA	1	0.018181818181818	0.06425935377650496	0.047284245674038744	0.3217665615141956	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.13943389023386946	22
DDC	1	0.018181818181818	0.06425935377650496	0.047284245674038744	0.28075709779179814	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.13328247067550983	23
GARS1	1	0.018181818181818	0.06425935377650496	0.047284245674038744	0.28075709779179814	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.13328247067550983	24
SEL1L	1	0.018181818181818	0.06425935377650496	0.047284245674038744	0.28075709779179814	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.13328247067550983	25
TNS3	1	0.018181818181818	0.06425935377650496	0.047284245674038744	0.28075709779179814	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.13328247067550983	26
PCDH15	1	0.018181818181818	0.06425935377650496	0.047284245674038744	0.27129337539432175	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.13186291231588837	27
VOPP1	1	0.018181818181818	0.06425935377650496	0.047284245674038744	0.2302839116719243	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.12571149275752877	28
SCAF4	1	0.018181818181818	0.06425935377650496	0.047284245674038744	0.11777076761303895	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.10883452114869596	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ZBPB	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.006309148264984188	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.09211527824648774	30
PLOD3	1	0.01818181818181818	0.06425935377650496	0.047284245674038744	0.0	0.6019267297326067	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.09116890600674012	31

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
30	1	40	0	14	None	1.0	-0.0
36	2	39	0	14	None	1.0	-0.0
142	3	38	0	14	None	0.5624166190203925	0.24994185469837954
249	3	38	1	13	1.0263157894736843	1.0	-0.0
288	3	38	2	12	0.47368421052631576	0.5924468510484049	0.22735060452753406
297	7	34	2	12	1.2352941176470589	1.0	-0.0
336	8	33	2	12	1.4545454545454546	1.0	-0.0
599	9	32	2	12	1.6875	0.7093311767774617	0.14915095146564436
627	9	32	3	11	1.03125	1.0	-0.0
640	10	31	3	11	1.1827956989247312	1.0	-0.0
651	11	30	3	11	1.3444444444444446	1.0	-0.0
687	11	30	6	8	0.4888888888888889	0.3219137899102455	0.49226041892280514
688	15	26	6	8	0.7692307692307693	0.7548366596998944	0.12214701587495107
823	15	26	7	7	0.5769230769230769	0.528760430668828	0.27674105235972185
875	15	26	8	6	0.4326923076923077	0.21856306955359578	0.6604232186170306
876	15	26	9	5	0.32051282051282054	0.1175979559898796	0.9296002268145651
981	15	26	10	4	0.23076923076923078	0.031971018585759146	1.4952435270336717
982	31	10	10	4	1.24	0.7358375514955705	0.1332180528854592
983	31	10	13	1	0.23846153846153847	0.25480492869908067	0.5937921756847577
986	32	9	13	1	0.27350427350427353	0.4229368950729017	0.37372442734387856
994	33	8	13	1	0.3173076923076923	0.4208384837822578	0.3758845527657329
1003	34	7	13	1	0.37362637362637363	0.663943317750199	0.17786899569082928
1038	35	6	13	1	0.44871794871794873	0.663943317750199	0.17786899569082928

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1054	38	3	13	1	0.9743589743589743	1.0	-0.0
1055	38	3	14	0	0.0	0.5624166190203926	0.24994185469837946
1060	39	2	14	0	0.0	1.0	-0.0
1075	40	1	14	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

6	13
22	14

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf01030	Receptor L domain	55	36	1	18	0.012328117591275486
auto_disruption_pf00757	Furin-like cysteine rich region	55	35	7	13	0.1103896103896104
auto_disruption_pf14843	Growth factor receptor domain IV	55	39	2	14	0.07623106060606061

domain_retention tables

frame_domain_contingency_table

25	16
3	11

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	55	41	3	11	0.10276679841897234

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
18	43	55	0.7818181818181819

frequency tables

Partner_gene_counts

Partner_gene	Event_count
BLTP3B	1
CADPS	1
CDH7	1
CSF2RA	1
DDC	1
EEA1	1
ELAPOR2	1
FADD	2
GARS1	1
KIF5B	1
LANCL2	3
NIPSNAP2	3
NUMA1	1
PCDH15	1
PDE1C	1
PKD1L1	1
PLOD3	1
RAD51	2
SCAF4	1
SEL1L	1
SEPTIN14	18
TNRC18	1
TNS3	1
TUT7	1
VOPP1	1
VPS41	1
VSTM2A	1
VWC2	2
YIF1B	1
ZNF713	1
ZPBP	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
7	55268000	55269000	23	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (23 total)
7	55241000	55242000	7	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (7 total)
7	55269000	55270000	5	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
7	55222000	55223000	4	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
7	55270000	55271000	3	EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, EVT-P-0042236-T01-IM6-530
7	55238000	55239000	2	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558
7	55259000	55260000	2	EVT-P-0030222-T01-IM6-557, EVT-P-0059222-T03-IM7-500
7	55179000	55180000	1	EVT-P-0059032-T01-IM7-546
7	55209000	55210000	1	EVT-P-0030222-T01-IM6-559
7	55214000	55215000	1	EVT-P-0028484-T02-IM6-548
7	55220000	55221000	1	EVT-P-0016009-T01-IM6-527
7	55221000	55222000	1	EVT-P-0063142-T01-IM7-503
7	55224000	55225000	1	EVT-P-0038683-T01-IM6-4
7	55233000	55234000	1	EVT-P-0059222-T01-IM7-540
7	55240000	55241000	1	EVT-P-0065470-T01-IM7-545
7	55254000	55255000	1	EVT-P-0064660-T01-IM7-9

exact_position_recurrence

chromosome	position	n_events	event_ids
7	55241077	2	EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535
7	55268444	2	EVT-P-0037759-T01-IM6-541, EVT-P-0037759-T03-IM6-542
7	55268799	2	EVT-P-0041149-T01-IM6-506, EVT-P-0041149-T04-IM6-505

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
982	16	15	700	False
297	4	4	281	False
688	4	4	476	False
687	3	2	432	False
983	3	2	93	False
1054	3	3	407	False

mechanistic_interpretation tables

retention_counter_intuitive_events

event_id	sample_id	partner_gene	breakpoint_protein_position	domain_status
EVT-P-0059222-T03-IM7-500	P-0059222-T03-IM7	LANCL2	875	disrupted
EVT-P-0029159-T01-IM6-516	P-0029159-T01-IM6	SEPTIN14	1055	lost
EVT-P-0016009-T01-IM6-527	P-0016009-T01-IM6	VOPP1	249	lost

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
97	297	200	7	48	5	36	0.8333333333333334	1.0	-0.0	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0028484-T02-IM6-548, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, ... (7 total)
136	336	200	8	47	6	35	1.0285714285714285	1.0	-0.0	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0028484-T02-IM6-548, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, ... (8 total)
142	342	200	8	47	6	35	1.0285714285714285	1.0	-0.0	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0028484-T02-IM6-548, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
197	297	100	6	49	4	37	0.6486486486486487	0.6380928037309834	0.19511615322676598	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
236	336	100	7	48	5	36	0.8333333333333334	1.0	-0.0	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, ... (7 total)
247	297	50	6	49	4	37	0.6486486486486487	0.6380928037309834	0.19511615322676598	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
249	299	50	6	49	4	37	0.6486486486486487	0.6380928037309834	0.19511615322676598	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
249	349	100	7	48	5	36	0.8333333333333334	1.0	-0.0	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, ... (7 total)
249	449	200	7	48	5	36	0.8333333333333334	1.0	-0.0	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, ... (7 total)
272	297	25	5	50	4	37	1.4054054054054055	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, EVT-P-0063743-T01-IM7-536
286	336	50	6	49	5	36	1.8055555555555556	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
288	313	25	5	50	4	37	1.4054054054054055	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, EVT-P-0063743-T01-IM7-536

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
288	338	50	6	49	5	36	1.805555555555556	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
288	388	100	6	49	5	36	1.805555555555556	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
288	488	200	6	49	5	36	1.805555555555556	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
297	322	25	4	51	4	37	None	0.5624166190203925	0.24994185469837954	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
297	347	50	5	50	5	36	None	0.3135946965025766	0.5036312906914963	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
297	397	100	5	50	5	36	None	0.3135946965025766	0.5036312906914963	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
297	497	200	5	50	5	36	None	0.3135946965025766	0.5036312906914963	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
451	651	200	4	51	3	38	1.0263157894736843	1.0	-0.0	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558, EVT-P-0059222-T01-IM7-540, EVT-P-0065470-T01-IM7-545
487	687	200	7	48	3	38	0.19736842105263158	0.06129697161883621	1.2125609813007856	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0039262-T01-IM6-529, ... (7 total)
488	688	200	11	44	7	34	0.5147058823529411	0.44288475549013484	0.35370926828253907	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, ... (11 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
551	651	100	4	51	3	38	1.0263157894736843	1.0	-0.0	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558, EVT-P-0059222-T01-IM7-540, EVT-P-0065470-T01-IM7-545
587	687	100	7	48	3	38	0.19736842105263158	0.06129697161883621	1.2125609813007856	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0039262-T01-IM6-529, ... (7 total)
588	688	100	11	44	7	34	0.5147058823529411	0.44288475549013484	0.35370926828253907	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, ... (11 total)
599	699	100	11	44	7	34	0.5147058823529411	0.44288475549013484	0.35370926828253907	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, ... (11 total)
599	799	200	11	44	7	34	0.5147058823529411	0.44288475549013484	0.35370926828253907	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, ... (11 total)
623	823	200	11	44	6	35	0.30857142857142855	0.12363130591147359	0.9078715433000231	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, ... (11 total)
627	727	100	10	45	6	35	0.42857142857142855	0.25524334321374503	0.5930455755612729	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, ... (10 total)
627	827	200	11	44	6	35	0.30857142857142855	0.12363130591147359	0.9078715433000231	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, ... (11 total)
637	687	50	5	50	2	39	0.18803418803418803	0.09817374634244778	1.008004635868936	EVT-P-0019922-T01-IM6-558, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0039262-T01-IM6-529, EVT-P-0065470-T01-IM7-545

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
638	688	50	9	46	6	35	0.6285714285714286	0.6782436021012543	0.1686142942640827	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, ... (9 total)
640	690	50	9	46	6	35	0.6285714285714286	0.6782436021012543	0.1686142942640827	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, ... (9 total)
640	740	100	9	46	6	35	0.6285714285714286	0.6782436021012543	0.1686142942640827	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, ... (9 total)
640	840	200	10	45	6	35	0.42857142857142855	0.25524334321374503	0.5930455755612729	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, ... (10 total)
651	701	50	8	47	5	36	0.5092592592592593	0.4054381775580443	0.39207535894312023	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (8 total)
651	751	100	8	47	5	36	0.5092592592592593	0.4054381775580443	0.39207535894312023	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (8 total)
651	851	200	9	46	5	36	0.3472222222222222	0.21045930054077028	0.6768318773082844	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (9 total)
663	688	25	7	48	4	37	0.3963963963963964	0.3537371495196134	0.4513193281293952	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (7 total)
675	875	200	9	46	4	37	0.1945945945945946	0.0373826447479835	1.4273299763968943	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (9 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
676	876	200	10	45	4	37	0.14414414414414414	0.011735395640112464	1.9305022642740601	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0030222-T01-IM6-557, ... (10 total)
687	712	25	7	48	4	37	0.3963963963963964	0.3537371495196134	0.4513193281293952	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (7 total)
687	737	50	7	48	4	37	0.3963963963963964	0.3537371495196134	0.4513193281293952	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (7 total)
687	787	100	7	48	4	37	0.3963963963963964	0.3537371495196134	0.4513193281293952	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (7 total)
687	887	200	10	45	4	37	0.14414414414414414	0.011735395640112464	1.9305022642740601	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0030222-T01-IM6-557, ... (10 total)
688	713	25	4	51	4	37	None	0.5624166190203925	0.24994185469837954	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3
688	738	50	4	51	4	37	None	0.5624166190203925	0.24994185469837954	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3
688	788	100	4	51	4	37	None	0.5624166190203925	0.24994185469837954	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3
688	888	200	7	48	4	37	0.3963963963963964	0.3537371495196134	0.4513193281293952	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0030222-T01-IM6-557, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3, ... (7 total)
781	981	200	4	51	0	41	0.0	0.002935010482180293	2.5323903433618757	EVT-P-0030222-T01-IM6-557, EVT-P-0034892-T01-IM6-497, EVT-P-0059222-T03-IM7-500, EVT-P-0064660-T01-IM7-9

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
782	982	200	20	35	16	25	1.6	0.5390091037092031	0.2684038996420494	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0028241-T01-IM6-517, ... (20 total)
783	983	200	23	32	16	25	0.64	0.5393172527040498	0.26815568645331866	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0025018-T01-IM6-539, ... (23 total)
786	986	200	24	31	17	24	0.7083333333333334	0.756109133784497	0.12141551563670067	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (24 total)
794	994	200	25	30	18	23	0.782608695652174	0.7619062002140677	0.11809849222591266	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (25 total)
803	1003	200	26	29	19	22	0.8636363636363636	1.0	-0.0	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (26 total)
823	1023	200	26	29	19	22	0.8636363636363636	1.0	-0.0	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (26 total)
838	1038	200	26	29	20	21	1.2698412698412698	0.7641737602120285	0.1168078789621499	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (26 total)
854	1054	200	29	26	23	18	1.7037037037037037	0.5371910044161063	0.2698712684626607	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (29 total)
855	1055	200	30	25	23	18	1.2777777777777777	0.7619062002140677	0.11809849222591266	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (30 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
860	1060	200	31	24	24	17	1.411764705882353	0.756109133784497	0.12141551563670067	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (31 total)
875	1075	200	32	23	25	16	1.5625	0.5393172527040498	0.26815568645331866	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (32 total)
876	1076	200	31	24	25	16	2.083333333333335	0.3502356120624568	0.4556396968137162	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (31 total)
882	982	100	17	38	16	25	8.32	0.04193876645023118	1.3773843476364434	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0028241-T01-IM6-517, ... (17 total)
883	983	100	20	35	16	25	1.6	0.5390091037092031	0.2684038996420494	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0025018-T01-IM6-539, ... (20 total)
886	986	100	21	34	17	24	1.7708333333333333	0.5285320378844122	0.27692868204681664	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (21 total)
891	1091	200	31	24	26	15	3.12	0.1175979559898796	0.9296002268145651	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (31 total)
894	994	100	22	33	18	23	1.9565217391304348	0.36140911369603457	0.4420008999820817	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (22 total)
903	1003	100	23	32	19	22	2.159090909090909	0.34977920810228746	0.4562060098225183	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (23 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
932	982	50	17	38	16	25	8.32	0.04193876645023118	1.3773843476364434	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0028241-T01-IM6-517, ... (17 total)
933	983	50	20	35	16	25	1.6	0.5390091037092031	0.2684038996420494	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0025018-T01-IM6-539, ... (20 total)
936	986	50	21	34	17	24	1.7708333333333333	0.5285320378844122	0.27692868204681664	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (21 total)
938	1038	100	24	31	20	21	2.380952380952381	0.22585127029255028	0.6461774624216327	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (24 total)
944	994	50	22	33	18	23	1.9565217391304348	0.36140911369603457	0.4420008999820817	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (22 total)
953	1003	50	23	32	19	22	2.159090909090909	0.34977920810228746	0.4562060098225183	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (23 total)
954	1054	100	27	28	23	18	3.1944444444444446	0.12115391842134521	0.916662535127397	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (27 total)
955	1055	100	28	27	23	18	2.3	0.22695752000537878	0.6440554227791042	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (28 total)
957	982	25	17	38	16	25	8.32	0.04193876645023118	1.3773843476364434	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0028241-T01-IM6-517, ... (17 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
958	983	25	20	35	16	25	1.6	0.5390091037092031	0.2684038996420494	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0025018-T01-IM6-539, ... (20 total)
960	1060	100	29	26	24	17	2.541176470588235	0.21540780911121526	0.6667385564120671	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (29 total)
961	986	25	21	34	17	24	1.7708333333333333	0.5285320378844122	0.27692868204681664	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (21 total)
969	994	25	22	33	18	23	1.9565217391304348	0.36140911369603457	0.4420008999820817	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (22 total)
975	1075	100	30	25	25	16	2.8125	0.12770417513209759	0.8937949037765996	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (30 total)
978	1003	25	23	32	19	22	2.159090909090909	0.34977920810228746	0.4562060098225183	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (23 total)
981	1006	25	23	32	19	22	2.159090909090909	0.34977920810228746	0.4562060098225183	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (23 total)
981	1031	50	23	32	19	22	2.159090909090909	0.34977920810228746	0.4562060098225183	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (23 total)
981	1081	100	30	25	25	16	2.8125	0.12770417513209759	0.8937949037765996	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (30 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
981	1181	200	31	24	26	15	3.12	0.1175979559898796	0.9296002268145651	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (31 total)
982	1007	25	22	33	19	22	3.1666666666666665	0.12453316871616134	0.9047149612438605	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (22 total)
982	1032	50	22	33	19	22	3.1666666666666665	0.12453316871616134	0.9047149612438605	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (22 total)
982	1082	100	29	26	25	16	3.90625	0.061394476785918806	1.2118706974140676	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (29 total)
982	1182	200	30	25	26	15	4.333333333333333	0.03197101858575915	1.4952435270336717	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (30 total)
983	1008	25	6	49	3	38	0.2894736842105263	0.16509826343344652	0.7822574947883147	EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0037526-T01-IM6-509, EVT-P-0037759-T01-IM6-541, ... (6 total)
983	1033	50	6	49	3	38	0.2894736842105263	0.16509826343344652	0.7822574947883147	EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0037526-T01-IM6-509, EVT-P-0037759-T01-IM6-541, ... (6 total)
983	1083	100	13	42	9	32	0.703125	0.7187948128823816	0.14339506559852147	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0025191-T02-IM6-518, ... (13 total)
983	1183	200	14	41	10	31	0.8064516129032258	0.7358375514955705	0.1332180528854592	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0025191-T02-IM6-518, ... (14 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
986	1086	100	10	45	9	32	3.65625	0.4229368950729018	0.3737244273438785	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, ... (10 total)
986	1186	200	11	44	10	31	4.193548387096774	0.25480492869908067	0.5937921756847577	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, ... (11 total)
991	1091	100	10	45	9	32	3.65625	0.4229368950729018	0.3737244273438785	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, ... (10 total)
994	1094	100	10	45	9	32	3.65625	0.4229368950729018	0.3737244273438785	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, ... (10 total)
994	1194	200	10	45	9	32	3.65625	0.4229368950729018	0.3737244273438785	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, ... (10 total)
1003	1103	100	9	46	8	33	3.1515151515151514	0.4208384837822578	0.3758845527657329	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, ... (9 total)
1003	1203	200	9	46	8	33	3.1515151515151514	0.4208384837822578	0.3758845527657329	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, ... (9 total)
1004	1054	50	4	51	4	37	None	0.5624166190203925	0.24994185469837954	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
1005	1055	50	5	50	4	37	1.4054054054054055	1.0	-0.0	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1010	1060	50	6	49	5	36	1.805555555555556	1.0	-0.0	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0042236-T01-IM6-530, EVT-P-0043097-T01-IM6-531, ... (6 total)
1025	1075	50	7	48	6	35	2.2285714285714286	0.6639433177501989	0.17786899569082937	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0042236-T01-IM6-530, ... (7 total)
1029	1054	25	4	51	4	37	None	0.5624166190203925	0.24994185469837954	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
1030	1055	25	5	50	4	37	1.4054054054054055	1.0	-0.0	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
1035	1060	25	6	49	5	36	1.805555555555556	1.0	-0.0	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0042236-T01-IM6-530, EVT-P-0043097-T01-IM6-531, ... (6 total)
1038	1063	25	6	49	5	36	1.805555555555556	1.0	-0.0	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0042236-T01-IM6-530, EVT-P-0043097-T01-IM6-531, ... (6 total)
1038	1088	50	7	48	6	35	2.2285714285714286	0.6639433177501989	0.17786899569082937	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0042236-T01-IM6-530, ... (7 total)
1038	1138	100	8	47	7	34	2.676470588235294	0.6639433177501988	0.17786899569082945	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, ... (8 total)
1038	1238	200	8	47	7	34	2.676470588235294	0.6639433177501988	0.17786899569082945	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1041	1091	50	7	48	6	35	2.2285714285714286	0.6639433177501989	0.17786899569082937	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, ... (7 total)
1050	1075	25	6	49	5	36	1.8055555555555556	1.0	-0.0	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0042236-T01-IM6-530, ... (6 total)
1054	1079	25	6	49	5	36	1.8055555555555556	1.0	-0.0	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0042236-T01-IM6-530, ... (6 total)
1054	1104	50	7	48	6	35	2.2285714285714286	0.6639433177501989	0.17786899569082937	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, ... (7 total)
1054	1154	100	7	48	6	35	2.2285714285714286	0.6639433177501989	0.17786899569082937	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, ... (7 total)
1054	1254	200	7	48	6	35	2.2285714285714286	0.6639433177501989	0.17786899569082937	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, ... (7 total)
1055	1105	50	4	51	3	38	1.0263157894736843	1.0	-0.0	EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, EVT-P-0042236-T01-IM6-530
1055	1155	100	4	51	3	38	1.0263157894736843	1.0	-0.0	EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, EVT-P-0042236-T01-IM6-530
1055	1255	200	4	51	3	38	1.0263157894736843	1.0	-0.0	EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, EVT-P-0042236-T01-IM6-530

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
781	981	200	4	51	0	41	0.0	0.002935010482180293	2.5323903433618757	EVT-P-0030222-T01-IM6-557, EVT-P-0034892-T01-IM6-497, EVT-P-0059222-T03-IM7-500, EVT-P-0064660-T01-IM7-9
676	876	200	10	45	4	37	0.14414414414414414	0.011735395640112464	1.9305022642740601	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0030222-T01-IM6-557, ... (10 total)
982	1182	200	30	25	26	15	4.333333333333333	0.03197101858575915	1.4952435270336717	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (30 total)
675	875	200	9	46	4	37	0.1945945945945946	0.0373826447479835	1.4273299763968943	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (9 total)
957	982	25	17	38	16	25	8.32	0.04193876645023118	1.3773843476364434	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0028241-T01-IM6-517, ... (17 total)
587	687	100	7	48	3	38	0.19736842105263158	0.06129697161883621	1.2125609813007856	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0039262-T01-IM6-529, ... (7 total)
982	1082	100	29	26	25	16	3.90625	0.061394476785918806	1.2118706974140676	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (29 total)
637	687	50	5	50	2	39	0.18803418803418803	0.09817374634244778	1.008004635868936	EVT-P-0019922-T01-IM6-558, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0039262-T01-IM6-529, EVT-P-0065470-T01-IM7-545
891	1091	200	31	24	26	15	3.12	0.1175979559898796	0.9296002268145651	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (31 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
954	1054	100	27	28	23	18	3.194444444444446	0.12115391842134521	0.916662535127397	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (27 total)
623	823	200	11	44	6	35	0.30857142857142855	0.12363130591147359	0.9078715433000231	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, ... (11 total)
982	1007	25	22	33	19	22	3.166666666666665	0.12453316871616134	0.9047149612438605	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (22 total)
975	1075	100	30	25	25	16	2.8125	0.12770417513209759	0.8937949037765996	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (30 total)
983	1008	25	6	49	3	38	0.2894736842105263	0.16509826343344652	0.7822574947883147	EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0037526-T01-IM6-509, EVT-P-0037759-T01-IM6-541, ... (6 total)
651	851	200	9	46	5	36	0.3472222222222222	0.21045930054077028	0.6768318773082844	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (9 total)
960	1060	100	29	26	24	17	2.541176470588235	0.21540780911121526	0.6667385564120671	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (29 total)
938	1038	100	24	31	20	21	2.380952380952381	0.22585127029255028	0.6461774624216327	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (24 total)
955	1055	100	28	27	23	18	2.3	0.22695752000537878	0.6440554227791042	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (28 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
986	1186	200	11	44	10	31	4.193548387096774	0.25480492869908067	0.5937921756847577	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, ... (11 total)
627	727	100	10	45	6	35	0.42857142857142855	0.25524334321374503	0.5930455755612729	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, ... (10 total)
640	840	200	10	45	6	35	0.42857142857142855	0.25524334321374503	0.5930455755612729	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, ... (10 total)
297	347	50	5	50	5	36	None	0.3135946965025766	0.5036312906914963	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
978	1003	25	23	32	19	22	2.159090909090909	0.34977920810228746	0.4562060098225183	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (23 total)
876	1076	200	31	24	25	16	2.0833333333333335	0.3502356120624568	0.4556396968137162	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (31 total)
663	688	25	7	48	4	37	0.3963963963963964	0.3537371495196134	0.4513193281293952	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (7 total)
688	888	200	7	48	4	37	0.3963963963963964	0.3537371495196134	0.4513193281293952	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0030222-T01-IM6-557, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3, ... (7 total)
969	994	25	22	33	18	23	1.9565217391304348	0.36140911369603457	0.4420008999820817	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (22 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
651	701	50	8	47	5	36	0.5092592592592593	0.4054381775580443	0.39207535894312023	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, EVT-P-0036028-T01-IM6-543, ... (8 total)
1003	1103	100	9	46	8	33	3.1515151515151514	0.4208384837822578	0.3758845527657329	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, ... (9 total)
986	1086	100	10	45	9	32	3.65625	0.4229368950729018	0.3737244273438785	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, ... (10 total)
991	1091	100	10	45	9	32	3.65625	0.4229368950729018	0.3737244273438785	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, ... (10 total)
588	688	100	11	44	7	34	0.5147058823529411	0.44288475549013484	0.35370926828253907	EVT-P-0006873-T01-IM5-554, EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, ... (11 total)
961	986	25	21	34	17	24	1.7708333333333333	0.5285320378844122	0.27692868204681664	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (21 total)
854	1054	200	29	26	23	18	1.7037037037037037	0.5371910044161063	0.2698712684626607	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (29 total)
958	983	25	20	35	16	25	1.6	0.5390091037092031	0.2684038996420494	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0025018-T01-IM6-539, ... (20 total)
782	982	200	20	35	16	25	1.6	0.5390091037092031	0.2684038996420494	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0028241-T01-IM6-517, ... (20 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
783	983	200	23	32	16	25	0.64	0.5393172527040498	0.26815568645331866	EVT-P-0015179-T01-IM6-525, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, EVT-P-0025018-T01-IM6-539, ... (23 total)
875	1075	200	32	23	25	16	1.5625	0.5393172527040498	0.26815568645331866	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (32 total)
297	322	25	4	51	4	37	None	0.5624166190203925	0.24994185469837954	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063743-T01-IM7-536
688	713	25	4	51	4	37	None	0.5624166190203925	0.24994185469837954	EVT-P-0018389-T02-IM6-6, EVT-P-0021038-T01-IM6-538, EVT-P-0036028-T01-IM6-543, EVT-P-0050679-T02-IM6-3
1029	1054	25	4	51	4	37	None	0.5624166190203925	0.24994185469837954	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526
247	297	50	6	49	4	37	0.6486486486486487	0.6380928037309834	0.19511615322676598	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
1038	1138	100	8	47	7	34	2.676470588235294	0.6639433177501988	0.17786899569082945	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, ... (8 total)
1025	1075	50	7	48	6	35	2.2285714285714286	0.6639433177501989	0.17786899569082937	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0042236-T01-IM6-530, ... (7 total)
1041	1091	50	7	48	6	35	2.2285714285714286	0.6639433177501989	0.17786899569082937	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, ... (7 total)
638	688	50	9	46	6	35	0.6285714285714286	0.6782436021012543	0.1686142942640827	EVT-P-0018389-T02-IM6-6, EVT-P-0019922-T01-IM6-558, EVT-P-0021038-T01-IM6-538, EVT-P-0027944-T01-IM6-534, EVT-P-0027944-T03-IM6-535, ... (9 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
983	1083	100	13	42	9	32	0.703125	0.7187948128823816	0.14339506559852147	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0025191-T02-IM6-518, ... (13 total)
983	1183	200	14	41	10	31	0.8064516129032258	0.7358375514955705	0.1332180528854592	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015419-T01-IM6-555, EVT-P-0025018-T01-IM6-539, EVT-P-0025191-T02-IM6-518, ... (14 total)
786	986	200	24	31	17	24	0.7083333333333334	0.756109133784497	0.12141551563670067	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (24 total)
860	1060	200	31	24	24	17	1.411764705882353	0.756109133784497	0.12141551563670067	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (31 total)
794	994	200	25	30	18	23	0.782608695652174	0.7619062002140677	0.11809849222591266	EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, EVT-P-0023043-T01-IM6-521, ... (25 total)
855	1055	200	30	25	23	18	1.2777777777777777	0.7619062002140677	0.11809849222591266	EVT-P-0005701-T01-IM5-514, EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, ... (30 total)
838	1038	200	26	29	20	21	1.2698412698412698	0.7641737602120285	0.1168078789621499	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (26 total)
272	297	25	5	50	4	37	1.4054054054054055	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, EVT-P-0063743-T01-IM7-536
1030	1055	25	5	50	4	37	1.4054054054054055	1.0	-0.0	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0043097-T01-IM6-531, EVT-P-0058375-T01-IM6-526

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1035	1060	25	6	49	5	36	1.805555555555556	1.0	-0.0	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0042236-T01-IM6-530, EVT-P-0043097-T01-IM6-531, ... (6 total)
1050	1075	25	6	49	5	36	1.805555555555556	1.0	-0.0	EVT-P-0005701-T01-IM5-514, EVT-P-0025191-T02-IM6-518, EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0042236-T01-IM6-530, ... (6 total)
286	336	50	6	49	5	36	1.805555555555556	1.0	-0.0	EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, EVT-P-0063142-T01-IM7-503, ... (6 total)
1055	1105	50	4	51	3	38	1.0263157894736843	1.0	-0.0	EVT-P-0029159-T01-IM6-516, EVT-P-0031404-T01-IM6-2, EVT-P-0039255-T02-IM6-510, EVT-P-0042236-T01-IM6-530
236	336	100	7	48	5	36	0.8333333333333334	1.0	-0.0	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, EVT-P-0052976-T02-IM7-552, ... (7 total)
551	651	100	4	51	3	38	1.0263157894736843	1.0	-0.0	EVT-P-0006873-T01-IM5-554, EVT-P-0019922-T01-IM6-558, EVT-P-0059222-T01-IM7-540, EVT-P-0065470-T01-IM7-545
97	297	200	7	48	5	36	0.8333333333333334	1.0	-0.0	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0028484-T02-IM6-548, EVT-P-0030910-T01-IM6-551, EVT-P-0052976-T02-IM7-552, ... (7 total)
136	336	200	8	47	6	35	1.0285714285714285	1.0	-0.0	EVT-P-0016009-T01-IM6-527, EVT-P-0016795-T01-IM6-5, EVT-P-0028484-T02-IM6-548, EVT-P-0030910-T01-IM6-551, EVT-P-0038683-T01-IM6-4, ... (8 total)
803	1003	200	26	29	19	22	0.8636363636363636	1.0	-0.0	EVT-P-0014220-T02-IM6-520, EVT-P-0015179-T01-IM6-525, EVT-P-0015419-T01-IM6-555, EVT-P-0016256-T01-IM6-524, EVT-P-0019603-T01-IM6-522, ... (26 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031404-T01-IM6-2	P-0031404-T01-IM6		Breast Cancer	BRCA	BLTP3B-EGFR	BLTP3B	unknown	five_prime	55270272	27	1075	False	1075	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - UHRF1BP1L (NM_015054) Rearrangement : t(7;12)(p11.2;q23.1)(chr7:g.55270272::chr12:g.100460508) Protein Fusion: mid-exon {EGFR:UHRF1BP1L}	NA
EVT-P-0050679-T02-IM6-3	P-0050679-T02-IM6		Germ Cell Tumor	NSGCT	CADPS-EGFR	CADPS	out-of-frame	three_prime	55241099	18	688	True	688	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		CADPS (NM_003716) - EGFR (NM_005228) fusion: t(3;7)(p14.2;p11.2)(chr3:g.62537410::chr7:g.55241099) Protein Fusion: out of frame {CADPS:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038683-T01-IM6-4	P-0038683-T01-IM6		Melanoma	ARMM	CSF2RA-EGFR	CSF2RA	unknown	three_prime	55224220	9	336	True	336	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Receptor L domain	Furin-like cysteine rich region	CSF2RA (NM_001161530)-EGFR (NM_005228) rearrangement: t(7;X)(p11.2;p22.33)(chr7:g.55224220::chrX:g.1401662) Protein Fusion: mid-exon {CSF2RA:EGFR}	NA
EVT-P-0016795-T01-IM6-5	P-0016795-T01-IM6		Breast Cancer	IDC	DDC-EGFR	DDC	out-of-frame	three_prime	55222204	8	297	True	297	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Receptor L domain	Furin-like cysteine rich region	DDC (NM_000790)-EGFR (NM_005228) Rearrangement: c.435+4059:DDC_c.889+359:EGFRinv Protein Fusion: out of frame {DDC:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018389-T02-IM6-6	P-0018389-T02-IM6		Colorectal Cancer	COAD	EEA1-EGFR	EEA1	in-frame	three_prime	55241575	18	688	True	688	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		EEA1 (NM_003566) - EGFR (NM_005228) rearrangement: t(7;12)(p11.2;q22)(chr7:g.55241575::chr12:g.932211089) Protein Fusion: in frame {EEA1:EGFR}	NA
EVT-P-0064660-T01-IM7-9	P-0064660-T01-IM7		Bladder Cancer	BLCA	EGFR-CDH7	CDH7	out-of-frame	five_prime	55254206	20	823	True	823	True	exact match	disrupted	0.43873517786561267	True	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		Protein tyrosine and serine/threonine kinase	EGFR (NM_005228) - CDH7 (NM_033646) rearrangement: t(7;18)(p11.2;q22.1)(chr7:g.55254206::chr18:g.63486938) Protein Fusion: out of frame {EGFR:CDH7}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034892-T01-IM6-497	P-0034892-T01-IM6		Glioma	GB	EGFR-ELAPOR2	ELAPOR2	unknown	three_prime	55268103	24	981	False	981	True	exact match	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		EGFR (NM_005228) rearrangement: c.2943:EGFR_chr7:g.86587163del Antisense Fusion	NA
EVT-P-0059222-T03-IM7-500	P-0059222-T03-IM7		Glioma	GB	EGFR-LANCL2	LANCL2	in-frame	five_prime	55259704	21	875	True	875	True	exact match	disrupted	0.6442687747035574	True	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		Protein tyrosine and serine/threonine kinase	EGFR (NM_005228) - LANCL2 (NM_018697) rearrangement: c.2625+137:EGFR_c.1008+3812:LANCL2del Protein Fusion: in frame {EGFR:L ANCL2}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057880-T01-IM6-501	P-0057880-T01-IM6		Glioma	GB	EGFR-LANCL2	LANCL2	in-frame	five_prime	55268871	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - LANCL2 (NM_018697) fusion: c. 2947-10: EGFR_c.205-8690:LANCL2del Protein Fusion: in frame {EGFR:L ANCL2}	NA
EVT-P-0063142-T01-IM7-503	P-0063142-T01-IM7		Renal Cell Carcinoma	CCRCC	EGFR-PCDH15	PCDH15	unknown	five_prime	55221820	7	288	False	288	True	exact match	lost	0.0	False	Receptor L domain	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Furin-like cysteine rich region	EGFR (NM_005228) - PCDH15 (NM_001142763) rearrangement: t(7;10)(p11.2;q21.1)(chr7:g.55221820::chr10:g.56187463) Protein Fusion: mid-exon {EGFR:PCDH15}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042236-T01-IM6-504	P-0042236-T01-IM6		Glioma	GB	EGFR-PKD1L1	PKD1L1	in-frame	five_prime	55268824	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - PKD1L1 (NM_138295) fusion: c. 2947-57:EGFR_c.8527-7825:PKD1L1inv Protein Fusion: in frame {EGFR:PKD1L1}	NA
EVT-P-0041149-T04-IM6-505	P-0041149-T04-IM6		Non-Small Cell Lung Cancer	LUAD	EGFR-RAD51	RAD51	in-frame	five_prime	55268799	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.55268799::chr15:g.40996189) Protein Fusion: in frame {EGFR:RAD51}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0041149-T01-IM6-506	P-0041149-T01-IM6		Non-Small Cell Lung Cancer	LUAD	EGFR-RAD51	RAD51	in-frame	five_prime	5526879	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.5526879::chr15:g.40996189) Protein Fusion: in frame {EGFR: RAD51}	NA
EVT-P-0034594-T01-IM6-507	P-0034594-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	5526854	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion : c.2947-110:EGFR_c.1120-1204:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038138-T01-IM6-508	P-0038138-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268381	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+275:EGFR_c.1119+2043:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0037526-T01-IM6-509	P-0037526-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	unknown	five_prime	55268916	25	994	False	994	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2982:EGFR_c.1119+1151:SEPT14inv Protein Fusion: mid-exon {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039255-T02-IM6-510	P-0039255-T02-IM6		Esophagogastric Cancer	STAD	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55270807	27	1091	True	1090	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3271+489:EGFR_c.818-3507:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0032604-T01-IM6-512	P-0032604-T01-IM6		Esophagogastric Cancer	ESCA	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268771	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-110:EGFR_c.1120-1204:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0031753-T01-IM6-513	P-0031753-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268412	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+306:EGFR_c.1119+2531:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0005701-T01-IM5-514	P-0005701-T01-IM5		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55269490	26	1054	True	1054	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15:EGFR_c.1119+2833:SEPT14inv Protein fusion: in frame (EGFR-SEPT14)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043525-T01-IM6-515	P-0043525-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268859	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0029159-T01-IM6-516	P-0029159-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	three_prime	55269850	27	1055	True	1055	True	exact match	lost	0.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			SEPT14 (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv Protein Fusion: in frame {SEPT14:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028241-T01-IM6-517	P-0028241-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268525	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2947-356:EGFR_c.1119+4312:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0025191-T02-IM6-518	P-0025191-T02-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55269542	26	1054	True	1054	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3162+67:EGFR_c.1120-396:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0061245-T01-IM7-519	P-0061245-T01-IM7		Ampullary Cancer	IAMPCA	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268815	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement: c.2947-66:EGFR_c.1119+2973:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0014220-T02-IM6-520	P-0014220-T02-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	unknown	five_prime	55268942	25	1003	False	1002	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-25 fused with SEPT14 exon 10): c.3008:EGFR_c.1120-4102:SEPT14inv Protein Fusion: mid-exon {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023043-T01-IM6-521	P-0023043-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268179	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion (EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+73:EGFR_c.1120-1180:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0019603-T01-IM6-522	P-0019603-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268171	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion ((EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+65:EGFR_c.1119+3306:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047356-T01-IM6-523	P-0047356-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268316	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c. 2946+210:EGFR_c.1120-2817:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-0016256-T01-IM6-524	P-0016256-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268390	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - SEPT14 (NM_207366) fusion(EGFR exons 1-24 fused to SEPT14 exons 9-10): c. 2946+284:EGFR_c.1120-3940:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015179-T01-IM6-525	P-0015179-T01-IM6		Glioma	GB	EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268650	24	982	True	982	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_002529) - SEPT14 (NM_014857) fusion (EGFR exons 1-24 fused with SEPT14 exon 10); c.2947-231:EGFR_c.1119+3910:SEPT14inv Protein fusion: in frame (EGFR-SEPT14)	NA
EVT-P-0058375-T01-IM6-526	P-0058375-T01-IM6		Endometrial Cancer	UMEC	EGFR-TNRC18	TNRC18	out-of-frame	five_prime	55269897	26	1054	True	1054	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - TNRC18 (NM_001080495) fusion: c.3163-313:EGFR_c.5720-6144:TNRC18inv Protein Fusion: out of frame (EGFR:TNRC18)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016009-T01-IM6-527	P-0016009-T01-IM6		Breast Cancer	IDC	EGFR-VOPP1	VOPP1	in-frame	five_prime	55220393	6	249	True	249	True	exact match	lost	0.0	False	Receptor L domain	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Furin-like cysteine rich region	EGFR (NM_005228) - VOPP1 (NM_030796); c.747+36:E GFR_c.55-23161 VOPP1in v Protein Fusion: in frame {EGFR:VOPP1}	NA
EVT-P-0039262-T01-IM6-529	P-0039262-T01-IM6		Glioma	GB	EGFR-VSTM2A	VSTM2A	out-of-frame	five_prime	55241509	17	687	True	687	True	exact match	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - VSTM2A (NM_182546) rearrangement: c.2062-105:EGFR_c.634+2190:VSTM2Adup Protein Fusion: out of frame {EGFR:VSTM2A}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-042236-T01-IM6-530	P-0042236-T01-IM6		Glioma	GB	EGFR-VWC2	VWC2	unknown	five_prime	55270225	27	1060	False	1059	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - VWC2 (NM_198570) fusion: c. 3178:EGFR_c.827-13914:VWC2dup Protein Fusion: mid-exon {EGFR:VWC2}	NA
EVT-P-043097-T01-IM6-531	P-0043097-T01-IM6		Glioma	GB	EGFR-ZNF713	ZNF713	in-frame	five_prime	55269400	25	1038	True	1038	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228) - ZNF713 (NM_182633) fusion: c. 3115-28:EGFR_c.49-962:ZNF713del Protein Fusion: in frame {EGFR:ZNF713}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027944-T01-IM6-534	P-0027944-T01-IM6		Non-Small Cell Lung Cancer	LNET	FADD-EGFR	FADD	unknown	five_prime	55241077	17	687	True	687	True	exact match	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330) Protein Fusion: mid-exon {EGFR:FADD}	NA
EVT-P-0027944-T03-IM6-535	P-0027944-T03-IM6		Small Cell Lung Cancer	CSCLC	FADD-EGFR	FADD	unknown	five_prime	55241077	17	687	True	687	True	exact match	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330) Protein Fusion: mid-exon {EGFR:FADD}	NA
EVT-P-0063743-T01-IM7-536	P-0063743-T01-IM7		Glioma	GB	GARS1-EGFR	GARS1	out-of-frame	three_prime	55222368	8	297	True	297	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Receptor L domain	Furin-like cysteine rich region	GARS (NM_002047) - EGFR (NM_005228) fusion: c. 1032-338:GARS_c.889+523:EGFRdel Protein Fusion: out of frame {GARS:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021038-T01-IM6-538	P-0021038-T01-I-M6		Non-Small Cell Lung Cancer	LUAD	KIF5B-EGFR	KIF5B	in-frame	three_prime	55241486	18	688	True	688	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		KIF5B (NM_004521) - EGFR (NM_005228) rearrangement: t(7;10)(p11.2;q11.22)(chr7:g.55241486::chr10:g.32312089) Protein Fusion: in frame {KIF5B:EGFR}	NA
EVT-P-0025018-T01-IM6-539	P-0025018-T01-I-M6		Non-Small Cell Lung Cancer	LUSC	LANCL2-EGFR	LANCL2	out-of-frame	three_prime	55268537	25	983	True	983	True	exact match	lost	0.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			LANCL2 (NM_018697) - EGFR (NM_005228) rearrangement: c.1259-255:LANCL2_c.2947-344:EGFR dup Protein Fusion: out of frame {LANCL2:EGFR}	NA
EVT-P-0059222-T01-IM7-540	P-0059222-T01-I-M7		Glioma	GB	NIPSNA P2-EGFR	NIPSNA P2	unknown	three_prime	55233046	15	599	False	599	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region	Growth factor receptor domain IV	GBAS (NM_001483) - EGFR (NM_005228) fusion: c.92+717:GBAS_c.1796:EGFRdup Protein Fusion: mid-exon {GBAS:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037759-T01-IM6-541	P-0037759-T01-IM6		Breast Cancer	IDC	NIPSNA P2-EGFR	NIPSNA P2	out-of-frame	three_prime	5526844	25	983	True	983	True	exact match	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		GBAS (NM_001483) - EGFR (NM_005228) Rearrangement : c.2946+338:EGFR_c.279-1159:GBASdup Protein Fusion: out of frame {GBAS:EGFR}	NA
EVT-P-0037759-T03-IM6-542	P-0037759-T03-IM6		Breast Cancer	BRCA	NIPSNA P2-EGFR	NIPSNA P2	out-of-frame	three_prime	5526844	25	983	True	983	True	exact match	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		GBAS (NM_001483) - EGFR (NM_005228) rearrangement: c.279-1159:GBAS_c.2946+338:EGFRdup Protein Fusion: out of frame {GBAS:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036028-T01-IM6-543	P-0036028-T01-IM6		Colorectal Cancer	COAD	NUMA1-EGFR	NUMA1	in-frame	three_prime	55241128	18	688	True	688	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		NUMA1 (NM_006185) - EGFR (NM_005228) rearrangement: t(7;11)(p11.2;q13.4)(chr7:g.55241128::chr11:g.71728415) Protein Fusion: in frame {NUMA1:EGFR}	NA
EVT-P-0065470-T01-IM7-545	P-0065470-T01-IM7		Glioma	GB	PDE1C-EGFR	PDE1C	unknown	three_prime	55240707	17	651	False	651	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		PDE1C (NM_001191058) - EGFR (NM_005228) fusion: c.86-15224:PDE1C_c.1951:EGFRin Protein Fusion: mid-exon {PDE1C:EGFR}	NA
EVT-P-0059032-T01-IM7-546	P-0059032-T01-IM7		Non-Small Cell Lung Cancer	LUAD	PLOD3-EGFR	PLOD3	out-of-frame	three_prime	55179084	2	30	True	30	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			PLOD3 (NM_001084) - EGFR (NM_005228) fusion: c.2062-9:PLOD3_c.89-30895:EGFRin Protein Fusion: out of frame {PLOD3:EGFR}	NA

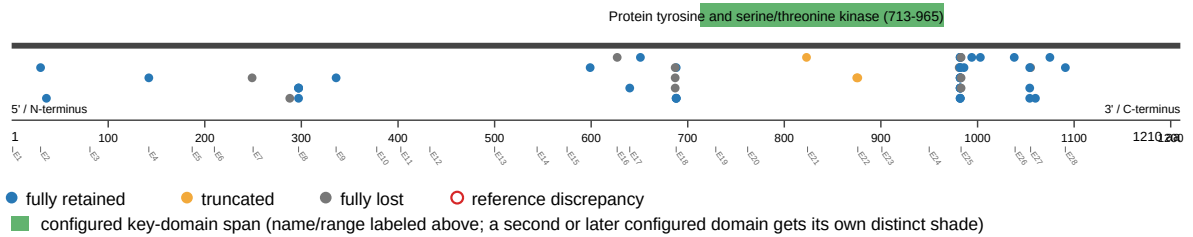
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028484-T02-IM6-548	P-0028484-T02-IM6		Bone Cancer	OS	SCAF4-EGFR	SCAF4	out-of-frame	three_prime	55214145	4	142	True	142	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Furin-like cysteine rich region; Growth factor receptor domain IV		Receptor L domain	SCAF4 (NM_020706) - EGFR (NM_005228) fusion: t(7;21)(p11.2;q22.11)(chr7:g.55214145::chr21:g.33094864) Protein Fusion: out of frame {SCAF4:EGFR}	NA
EVT-P-0030910-T01-IM6-551	P-0030910-T01-IM6		Pancreatic Cancer	PAAC	SEL1L-EGFR	SEL1L	in-frame	three_prime	55222485	8	297	True	297	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Receptor L domain	Furin-like cysteine rich region	SEL1L (NM_005065) - EGFR (NM_005228) rearrangement: t(7;14)(p11.2;q31.1)(chr7:g.5522485::chr14:g.81986576) Protein Fusion: in frame {SEL1L:EGFR}	NA
EVT-P-0052976-T02-IM7-552	P-0052976-T02-IM7		Glioma	GB	TNS3-EGFR	TNS3	in-frame	three_prime	55222307	8	297	True	297	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Growth factor receptor domain IV	Receptor L domain	Furin-like cysteine rich region	TNS3 (NM_022748) - EGFR (NM_005228) fusion: c.1025-206:TNS3_c.889+462:EGFR Protein Fusion: in frame {TNS3:EGFR}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006873-T01-IM5-554	P-0006873-T01-IM5		Non-Small Cell Lung Cancer	LUSC	TUT7-EGFR	TUT7	unknown	five_prime	55238031	15	627	True	626	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Receptor L domain; Furin-like cysteine rich region	Protein tyrosine and serine/threonine kinase	Growth factor receptor domain IV	EGFR (NM_005228) - ZCCHC6 (NM_024617) rearrangement : t(7;9)(chr7p11.2;chr9q21.33)(chr7:g.55238031::chr9:g.88930920) Protein fusion: mid-exon (EGFR-ZCCHC6)	NA
EVT-P-0015419-T01-IM6-555	P-0015419-T01-IM6		Nerve Sheath Tumor	MPNST	VPS41-EGFR	VPS41	unknown	five_prime	55268890	25	986	False	985	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			EGFR (NM_005228)-VPS41 (NM_014396) Rearrangement : c.2956:EGFR_c.247-4922:VPS41inv Protein fusion: mid-exon (EGFR-VPS41)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030222-T01-IM6-557	P-0030222-T01-IM6		Glioma	GB	VWC2-EGFR	VWC2	out-of-frame	three_prime	55259621	22	876	True	876	True	exact match	disrupted	0.3557312252964427	True		Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV	Protein tyrosine and serine/threonine kinase	VWC2 (NM_198570) - EGFR (NM_005228) rearrangement: c.827-37789:VWC2_c.2625+54:EGFRdel Protein Fusion: out of frame {VWC2:EGFR}	NA
EVT-P-0019922-T01-IM6-558	P-0019922-T01-IM6		Soft Tissue Sarcoma	DDLS	YIF1B-EGFR	YIF1B	unknown	three_prime	55238939	17	640	True	640	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV		EGFR (NM_005228) rearrangement: t(7;19)(p22.1;q13.2)(chr7:g.5523893::chr19:g.38806683) Antisense Fusion	NA
EVT-P-0030222-T01-IM6-559	P-0030222-T01-IM6		Glioma	GB	ZPBP-EGFR	ZPBP	unknown	three_prime	55209996	2	36	False	36	True	exact match	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Receptor L domain; Furin-like cysteine rich region; Growth factor receptor domain IV			ZPBP (NM_007009) - EGFR (NM_005228) rearrangement: c.487+5719:ZPBP_c.106:EGFRinv Protein Fusion: mid-exon {ZPBP:EGFR}	NA

domain retention outliers

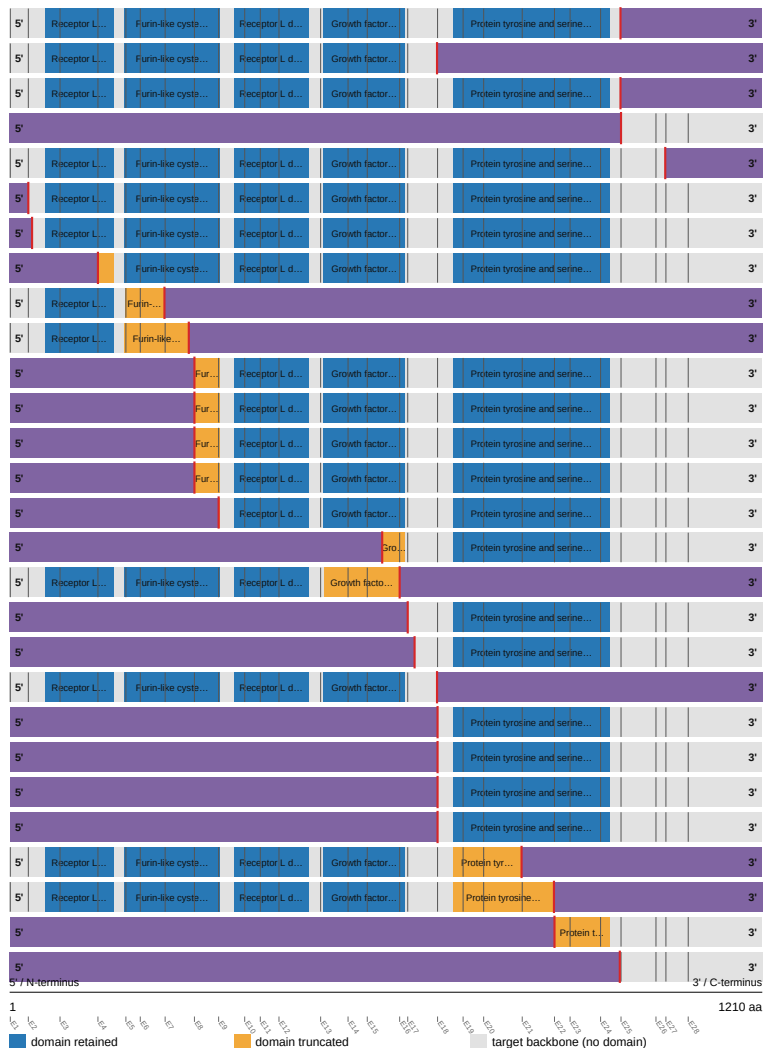
EGFR domain-retention track



fusion schematic

EGFR fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (EGFR portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Showing the top 28 of 40 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

SEPTIN14 (x12) – Protein tyrosine and serine/threonine kinase retained – Glioma, Ampullary Cancer, E

FADD (x2) – Protein tyrosine and serine/threonine kinase lost – Non-Small Cell Lung Cancer, Small Cell

RAD51 (x2) – Protein tyrosine and serine/threonine kinase retained – Non-Small Cell Lung Cancer

NIPSNAP2 (x2) – Protein tyrosine and serine/threonine kinase lost – Breast Cancer

SEPTIN14 (x2) – Protein tyrosine and serine/threonine kinase retained – Glioma

PLOD3 – Protein tyrosine and serine/threonine kinase retained – Non-Small Cell Lung Cancer

ZBPB – Protein tyrosine and serine/threonine kinase retained – Glioma

SCAF4 – Protein tyrosine and serine/threonine kinase retained – Bone Cancer

VOPP1 – Protein tyrosine and serine/threonine kinase lost – Breast Cancer

PCDH15 – Protein tyrosine and serine/threonine kinase lost – Renal Cell Carcinoma

DDC – Protein tyrosine and serine/threonine kinase retained – Breast Cancer

GARS1 – Protein tyrosine and serine/threonine kinase retained – Glioma

SEL1L – Protein tyrosine and serine/threonine kinase retained – Pancreatic Cancer

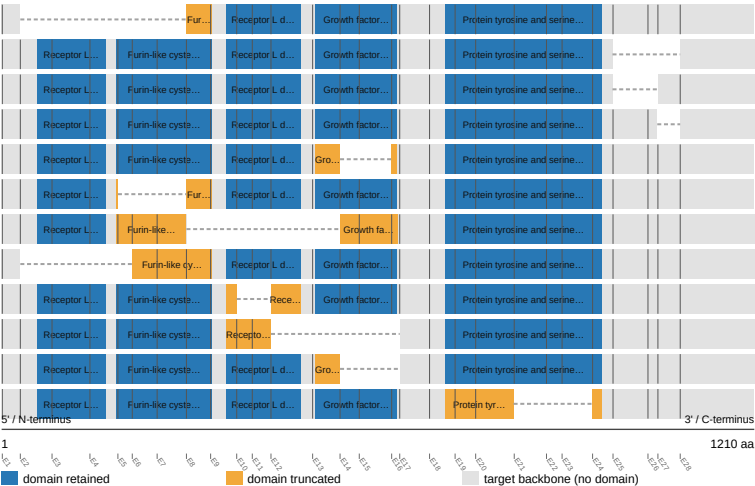
TNS3 – Protein tyrosine and serine/threonine kinase retained – Glioma

ELAPOR2 – Protein tyrosine and serine/threonine kinase lost – Glioma

intragenic deletion schematic

EGFR intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



- 6-exon deletion (in-frame) (x279)
- 3-exon deletion (out-of-frame) (x43)
- 2-exon deletion (in-frame) (x10)
- 1-exon deletion (out-of-frame) (x8)
- 2-exon deletion (in-frame) (x5)
- 3-exon deletion (in-frame) (x2)
- 6-exon deletion (out-of-frame) (x2)
- 4-exon deletion (in-frame)
- 2-exon deletion (in-frame)
- 5-exon deletion (in-frame)
- 3-exon deletion (in-frame)
- 3-exon deletion (out-of-frame)

breakpoint